

# Would Users Accept Electric Muscle Stimulation Controlling their Body? Insights from a Scenario-based Investigation

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**Figure 1: Illustration of the four EMS-based scenarios (1) motor learning, (2) virtual reality, (3) media player control, and (4) pedestrian safety (from left to right).**

## ABSTRACT

Electric muscle stimulation (EMS) offers rich opportunities for haptic interaction. However, there is no comprehensive study assessing the user acceptance of this technology. Consequently, we synthesized four scenarios (motor learning, virtual reality, media player control, and pedestrian safety) based on various dimensions identified by the literature review. We assessed their user acceptance in an online survey ( $N=113$ ) using the technology acceptance model (TAM). Our findings suggest that users may reject EMS systems if they perceive a high level of risk and lack of control over the environment. This was evident even in scenarios that could provide high benefits to users, such as accelerating the learning curve in motoric tasks. In contrast, EMS seems to be widely accepted for games and entertainment, represented by the virtual reality scenario in our study. HCI researchers should employ EMS to create more engaging VR experiences, keep users in control, and provide them the option of learning skills with or without EMS. Our results will help build better EMS systems accepted by users.

## CCS CONCEPTS

• **Human-centered computing** → **User studies; Haptic devices.**

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## KEYWORDS

Human actuation, Electrical muscle stimulation, Human-computer interaction

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## 1 INTRODUCTION

Electrical muscle stimulation (EMS) is a technology that can activate and control bodily movements with electrical signals. While EMS has been intensively researched in medicine, genetics, biomechanics, and psychology, the human-computer interaction (HCI) community has recently begun to gain interest into the technology. Prototypes using EMS have been proposed for applications such as motor learning [42], force feedback [26], conveying object affordances [24], pedestrian walking/navigation support [33], or assisted drawing [27]. EMS is often considered as a haptic technology [25], since EMS electrodes are typically placed on human skin (which is not a necessary prerequisite, EMS itself would also work hypodermic). Still, EMS goes beyond: due to the capability to initiate muscle contraction it allows to feed machine output directly into human motor control. This is a unique property, which extends the possibilities of “human in the loop” systems. For instance, Lopes et al. [24] utilized EMS to prevent users from burning their hand at a cup of hot water, by simply flexing the wrist away. Another example is the work by Schneegass and Rzayev [38], who twisted the arms of users to make them look at their watches to remind them

of upcoming meetings. Automation utilizing such interfaces can “skip” human information processing stages (such as the perceptual level) by directly entering the action stage.

However, this property could also become a major hurdle for a wide-ranging adoption of EMS technology in human-computer interaction. By bypassing the cognitive system, EMS interfaces may limit users’ sense of agency (i.e., their feeling of being in control), revealing ethical and even philosophical issues. Controllability is currently a highly discussed topic in multiple domains such as human-AI or human-automation interaction. For example, it has shown being a major barrier for the societal adoption of automated driving systems [9], [37]. Consequently, when thinking of EMS based interfaces, a highly relevant question is: *would users accept a technology that controls their own body?*

Existing work has typically been evaluated by showing its benefits on users’ performance (e.g., [6], [42]), demonstrating user understanding of stimulated movements (e.g., [34], [13]), showing that users can mimic movements generated by EMS (e.g., [11], [28]), or by delighting users in novel experiences [20], [26]. Some of these studies have briefly reported subjective comments from users, which clearly indicate doubts, for example “*I felt like my forearm was pushed by someone.*”, “*I felt like when I got a cramp in my arm.*” or “*I felt like my body was hacked.*” [42]. In this regard, a systematic assessment of EMS technology from the perspective of user acceptance is, to our best knowledge, still missing.

With this paper we aimed at contributing to these issues by assessing the user acceptance of different scenarios portraying potential EMS applications. By literature review, we identified several dimensions which can be used to categorize the use of this technology, and we used these dimensions to describe four scenarios based on existing prototypes. The scenarios were presented to participants in the form of an online survey with textual descriptions and images. For each scenario, study participants completed a version of the technology acceptance model (TAM) with trust integration [43], [44], [2]. With the results of this experiment, we want to contribute to this line of research by identifying potential pitfalls and opportunities for the design of novel EMS systems and prototypes in HCI. We support EMS designers with following guidelines for systems design: a) Leverage EMS in VR to create more engaging and immersive user experiences, b) The use of EMS to speed up learning was not widely embraced, emphasizing the importance of providing individuals a choice, c) Keep users in control of their activities rather than forcing them through EMS.

## 2 RELATED WORK

In this section, we introduce the construct of user acceptance, present a brief review of EMS systems proposed in the domain of HCI, and derive dimensions to categorize these systems with respect to our research goals.

### 2.1 User Acceptance as Precursor for the Use of Technology

User acceptance is one of the most important factors in deciding whether technology succeeds or fails. Davis [4] proposed the *Technology Acceptance Model* (TAM), which is based on the Psychological Attitude Paradigm [7] linking users’ values, attitudes, and (potential) behaviors. In the TAM, the perceived usefulness (PU) and ease of use (PeOU) of an external stimulus

or design feature influence the general attitude towards using a device (ATT, affective response), which in turn contributes to either use or disuse of a system (Intent, i.e., the behavioral response). The TAM has received various updates over the years, leading to TAM 2 [43] and TAM 3 [44]. Since trust in automation is an important additional factor, we utilized a TAM extension proposed by Belanche et al. [2]. They showed that both attitude and purpose to use are strongly linked to trust, making it a third belief “*able to explain the mechanism*”. TAM possesses the following qualities: parsimony (traits of being simple or having fewer variables), verifiability (data support) and generalizability (theory’s use in numerous disciplines) [3]. It was discovered to be a model that better explains the behavior of those who may be positively inclined toward technology [29], hence it was chosen as the study’s foundation.

### 2.2 Electric Muscle Stimulation

**2.2.1 Theory.** EMS mimics the body’s own electrical signals by inducing electrical current to the muscles, which can generate muscle contraction and, as a consequence, body movements. Typical EMS systems use a pair of electrodes (a cathode and an anode) to deliver such electrical signals [18]. The current passes through the nerves and motor neurons when traveling from cathode to anode, which increases the internal voltage of the neurons. This causes the motor neurons to “fire” when a certain threshold is exceeded [18], [32], thereby contracting the muscles associated with these neurons. The power needed to actuate a muscle varies between body parts and also between users, which requires a calibration process to ensure pain-free operation [17].

**2.2.2 EMS in HCI.** EMS provides many opportunities for novel interaction scenarios. By a variation of stimulus parameters (i.e., amplitudes, pulse widths, frequencies) it can be used to communicate object affordances [24], initiate direct user movements [21] [23], or to provide haptic feedback [28]. One of the first demonstrations in the context of HCI was “PossessedHand” [42], which manipulated a user’s fingers. Affordance++ [24] stimulated the arm of users to inform them the way they should communicate with an object. Lopes et al. [25] has proposed to use EMS in virtual reality environments. Impacto used a mixture of tactile feedback (a solenoid tapping the skin) and EMS to mimic the feeling of hitting or being hit in VR boxing [22]. In an augmented reality tennis game, Jain et al. [14] used EMS on the wrist muscles to simulate the feeling of a ball hitting a racket. Kasahara et al. [15] have suggested to use the technology to speed up human reaction times. They showed that with proper timing of signal stimulation, objects can be grabbed faster than usual. Pose-IO offered both input and output based on proprioception sense using EMS [23]. Users entered data by flexing their wrist in an input motion, which the device detected using a 3-axis accelerometer. Pose-IO output was triggered when a user’s wrist was positioned in an output gesture, which Pose-IO actuated utilizing EMS [23].

EMS was also used for learning/ training purposes. For example, Ebisu et al. [5] manipulated users’ movements in order to learn playing percussion by simulating both arms and legs. Despite no expert percussionists were involved in this process, they demonstrated how students could acquire the rhythm. Students could learn not only the rhythm with one hand, but also the rhythm with both hands at the same time. Furthermore, it was feasible to learn to play drum beats that demand a musician to use both the arms and the feet.

Also, pedestrian navigation with EMS has been proposed. Pfeiffer et al. [33] guided a pedestrians’ walking course, and “FootStriker” sensed heel striking when running and corrected it in real-time to mid-/forefoot running by applying EMS to one of the calf muscles [11]. However, the utilization of EMS is not limited to humans’ limbs. Nijima and Ogawa [30] stimulated face muscles to create food textures when eating, others have used EMS to make users smile [45], and even a stimulation of abdominal a laryngeal muscle was suggested as a potential method to assist singers in a vibrato [10]. Finally,

some researchers used EMS to investigate embodied emotional feedback by combining the actuation of the human body with the stimulation of perceptions [12].

Some of the aforementioned experiments have assessed users' perceptions, yielding to mixed results in terms of user acceptance. For example, participants of Pohl et al. [35] study detested the greater temporal rhythms of EMS as the signal amplitude increased. Also, in study [35], [36] participants have responded EMS leading to a "a weird feeling" when they "can't control their muscles". Control perceptions appear to be important for prospective users [40]. This lack of autonomy is a weakness that researchers must address. However, recent efforts from our community to address the issue of agency have been promising. Kasahara et al. [15] addressed the loss of agency in EMS by delaying the muscle stimulation signal to get it closer to the user's own reaction time while still accelerating the user's reaction time. Tajima et al. [41] created a framework that allows digital-touch system researchers and practitioners to prioritize/trade-off between haptic-assistance and sense-of-agency. Consequently, we have reason to doubt that potential users will blindly accept and use EMS systems in their daily lives.

**2.2.3 EMS Taxonomy: Not all systems are equal.** A closer examination of the various EMS prototypes described above suggests that they may be classified along different dimensions. We believe that categorizing data in this way would make it easier to compare and generalize results achieved with a certain prototype. Based on the literature, we came up with the following list:

- **Controllability:** This dimension refers to a users' ability of being in control of the situation (i.e., the "sense of agency"). For example, when using EMS in VR [25] the feedback generated by the system reflects the behavior of the user in the simulated environment (i.e., *high controllability*). In contrast, an EMS system making users to do certain movements (such as smile [30] or change walking direction [33]) without them intending to do so indicates *low controllability*. One important remark here is that control requires additional input modalities, such as VR controllers or "electric muscle reading".
- **Intrusiveness:** This category describes how intriguing an EMS interface behaves. For example, *high intrusiveness* is present when a system excessively controls your body parts (such as [42]). Other prototypes, such as [34] may intercede only slightly and infrequently with the user.
- **Longevity:** An EMS interface may stimulate muscles repeatedly (for example, [27]), while other systems (such as a system preventing to grab a hot mug [24]) trigger users only in certain events and for a limited amount of time.
- **Safety-criticality:** Further, the coffee mug prototype is an example of EMS being used to prevent a safety-critical situation [24]. Also, pedestrian navigation [33] operates in a potentially safety-critical environment. Other environments (such as VR feedback) where EMS is used are less critical.
- **Open vs. closed loop:** Some prototypes have combined muscle reading and muscle writing (electromyography with EMS) [13] or other input techniques (like VR-controller). Such systems do not only provide input to the user, they also allow users to control their limbs to provide input to the system. Although this is not a property of EMS systems per se, we argue that users would not necessarily see this technical difference.

### 3 METHOD AND RESEARCH QUESTIONS

We derived four EMS-based interaction scenarios based on our literature study that varied along the aspects outlined above (see Figure 1 and Table 1). Although there are many more EMS scenarios, particularly those connected

to rehabilitation ([16], [39]), we focused on possible daily uses for a broad audience rather than specialist ones:

- **Motor Learning:** EMS could be a promising technology to fasten the learning progress for motoric tasks (see [42], [5]). We used the example of guitar playing, where a user could be supported by EMS during a motor skill building activity (i.e., automatically following the guitar tabs).
- **Virtual Objects:** In this scenario, EMS supported immersion in virtual reality (VR) environments by providing situation-specific haptic feedback (see [25] [26]). For example, when a user picked a chair in VR, s/he could feel the resistance of holding the object.
- **Media Player:** EMS may support classical interactions such as substituting screen controls. We used the example of a Media Player (see [23]), where the current progress of the video is reflected on the user's body, while at the same time the user is able to perform actions (such as rewind/forward the video). The hand position is mapped to the timeline of the video - a straight wrist marking the beginning of the video, a fully flexed wrist marking the ending.
- **Road Safety:** As EMS was already used for pedestrian navigation (see [33]), we hypothesized how this could be extended to increase pedestrian safety. We described a scenario where a user would be held back by the system in critical situations, for example, when stepping onto the street when a fast car is approaching.

Those scenarios represent different manifestations of our proposed EMS categories (see Table 1). For instance, in the Motor Learning and Media Player scenarios, EMS signals control user movements for longer period of time, compared to the Virtual Objects and Road Safety scenario, which operate event based. Similarly, in the Motor Learning and Road Safety scenarios controllability is lower compared to the other two scenarios. The Road Safety scenario is the only safety-critical setting. When it comes to interaction directionality (muscle reading and muscle writing), Media Player scenarios include both.

We evaluated the user acceptance of these scenarios in an online survey using the TAM model with trust extension [2, 43, 44]. We provided the description of the four scenarios along with a reference image to aid with the imagination. All study participants rated all four scenarios in a within-subjects design, and by directly comparing them in terms of user acceptance. We aimed at answering the following research question:

**Which EMS interaction scenarios are accepted by users, and can the findings be related to system properties/dimensions?**

## 4 ONLINE SURVEY

### 4.1 Scenario descriptions

Participants were given the following application scenarios for EMS systems and instructed to visualize themselves utilizing the technology while answering survey questions.

**4.1.1 Motor Learning:** *Imagine you want to learn a new musical instrument, let's say: a guitar. Learning guitar from scratch takes several years until you can control your fingers in the necessary way. However, there is a much faster way to learn motor skills: you put an EMS system onto your hand, which then stimulates your finger muscles using continuous electric signal during the training session. So, you can even play your first song already in the first training session. Potentially learning new tasks will become much easier using EMS systems. In this scenario, you would constantly feel the EMS operating your muscles while playing the instrument. Imagine yourself learning a new instrument using this EMS technology.*

**4.1.2 Virtual Objects:** *Current virtual reality environments lack the haptics dimension. When you lift an object in a virtual environment, it feels unreal and disengaging as no object weight is experienced. Imagine the next step towards realism and immersion, EMS system support the haptic sense in order*

| Scenario        | Controllability | Intrusiveness | Longevity | Safety-criticality | Open vs. closed loop |
|-----------------|-----------------|---------------|-----------|--------------------|----------------------|
| Motor Learning  | Low             | High          | High      | Low                | Open                 |
| Virtual Objects | High            | Low           | Low       | Low                | Open                 |
| Media player    | High            | High          | High      | Low                | Closed               |
| Road safety     | Low             | High          | Low       | High               | Open                 |

**Table 1: This table shows the different dimensions along which we categorized EMS- based systems. This list was elaborated by reviewing recent literature on the topic.**

to convey the physicality of the virtual world. If in virtual environment you would try to push an object away, at that instant EMS system would create a counter force that would pull your arm backwards. This can significantly enhance the experience and can make it more fun. In this scenario, you would feel the EMS in certain, limited situations, for example when moving around an object. Imagine yourself playing a game in VR using this EMS technology.

**4.1.3 Media Player:** Imagine interacting with a device through the pose of your body rather than clicking on a button or talking to your phone. EMS system would understand your gestural command and perform the needed action, for instance, you can control video playback using your wrist. EMS system can tightly couple position of your wrist with the position of the video play head. You can set the position of the play head by posing your wrist. In this scenario, you would constantly feel the EMS operating your muscles while interacting.

**4.1.4 Road Safety:** Traffic safety is an important issue, and the number of pedestrians involved in accidents has increase over the last years. Often, pedestrians are overrun as they do not pay sufficient attention to the road environment. EMS systems may be able to improve pedestrian safety, as preemptive feedback can speed up human reaction time. Imagine yourself wearing an EMS device that senses nearby vehicles and preemptively stops you stepping on the street in critical situations (i.e., a fast car approaching) to ensure your safety by sending electrical signal. You might feel a buzz and, for a short moment, not being in control of your body. However, the system prevents you from stepping on the road and being hit by a car. In this scenario, you would feel the EMS operating your muscles only in short, safety-critical situations.

## 4.2 Participants and Procedure

The survey was set up and conducted in English. We relied on random sampling technique for the selection of survey participants. We distributed the survey on university mailing lists and on social media. In total 113 respondents (77 males, 31 females, 3 diverse, and 3 preferred not to provide details) participated in the study. The mean age of the participants was 28.69 (SD = 7.31) years. 77.2 % of the participants had no prior experience with EMS, with the rest having only done so in the context of medical applications or pain treatment. After completing the consent form, participants were presented a short comic-book like sketch describing what EMS technology is and how it works. Then, the four scenarios (descriptions and image) were presented in randomized order. We emphasized each scenario to be technically sophisticated and error-free. Consequently, they were asked to score their acceptance based on their personal feelings about these situations, rather than on their doubts about whether one or the other could be supported by technology. They completed the survey questions rating their user acceptability after each scenario. At the end of the survey, participants completed additional questions about demographics and their technology affinity (see below). The duration of the study was approximately 30 minutes per participant.

## 4.3 Measurements

To assess user acceptance, participants completed the TAM with trust extension based on Belanche et al. [2]. The scale reliability according to Cronbach's  $\alpha$  is acceptable for all sub-scales (around .7). The model includes the perceived usefulness (PU), perceived ease of use (PEoU), attitude towards using the system (ATT), safety and trust in automation (trust), as well as intention to use the system (Int). Consequently, we built scale values of all TAM dimensions for evaluation (see Table 2). Further, we included the affinity for technology interaction questionnaire ([1], [8]) to find out if acceptance towards EMS technology is related to technology affinity. This scale has 9 items, which can be averaged to a single overall score. The TAM items were assessed on a 7-point Likert scale (1 = "Completely Disagree" to 7 = "Completely Agree"), the ATI scale on a 6-point Likert scale, respectively (no neutral item). Finally, study participants could express their opinion towards EMS technology using a free text field at the end of the study. By that, we also wanted to get some qualitative statements.

## 4.4 RESULTS

In the following, we present a detailed evaluation of the obtained results. Statistical comparisons were carried out using IBM SPSS Version 27 and results are reported as statistically significant at  $p < .05$ . We conducted non-parametric Friedman ANOVAs with subsequent pairwise comparisons (Post-hoc Wilcoxon signed rank tests). To account for multiple tests, we applied Bonferroni correction with respect to the number of TAM-subcales (see Table 2). Descriptive statistics can be seen in Table 3.

| TAM (Bonferroni $\alpha = 0.01$ ) | Items | Cronbach's $\alpha$ |
|-----------------------------------|-------|---------------------|
| Perceived usefulness (PU)         | 3     | 0.698               |
| Perceived ease of use (PEoU)      | 5     | 0.730               |
| Attitude (Att)                    | 5     | 0.694               |
| Trust and Safety (trust)          | 4     | 0.778               |
| Intention (Int)                   | 1     | NA                  |

**Table 2: Utilized TAM dimensions with the respective number of questionnaire items and scale reliability (Cronbach's  $\alpha$ ).**

**4.4.1 Perceived usefulness (PU).** : There was a statistically significant difference in PU depending on the scenarios ( $\chi^2(2) = 86.294, p < .001$ ). Post-hoc test showed that the Road Safety scenario was perceived as significantly less useful than all the three other scenarios Motor Learning ( $Z = -7.075, p < .001$ ), VR ( $Z = -7.385, p < .001$ ), and Media Player ( $Z = -7.066, p < .001$ ). No other pairwise comparisons yielded significant differences.

**4.4.2 Perceived ease of use (PEoU).** : We obtained a statistically significant difference based on the scenarios ( $\chi^2(2) = 50.328, p < .000$ ). Again, the Road Safety scenario was rated significantly worse than all other scenarios Motor Learning ( $Z = -6.257, p < .001$ ), Virtual Objects ( $Z = -5.152, p < .001$ ), and Media Player ( $Z = -5.137, p < .001$ ). Also, there were again no differences between the latter three.

**4.4.3 Attitude (Att).** : Friedman ANOVA indicated a significant difference in attitude depending on scenarios, ( $\chi^2(2) = 92.852, p < .001$ ). Just as above, scenario Road Safety was rated significantly worse than Motor Learning ( $Z = -6.534, p < .001$ ), Virtual Objects ( $Z = -8.109, p < .001$ ), and Media Player ( $Z = -6.920, p < .001$ ). However, here we obtained additional differences as well: The Virtual Objects scenario was rated best, significantly better than the scenarios Motor Learning ( $Z = -4.604, p < .001$ ) and Media Player ( $Z = -3.196, p < .001$ ).

**4.4.4 Trust and Safety (trust).** : We found statistically significant difference based on the scenarios ( $\chi^2(2) = 63.740, p < .001$ ). As in all sub-scales, the Road Safety scenario was perceived being significantly less trustworthy than the other scenarios Motor Learning ( $Z = -6.039, p < .001$ ), Virtual Objects ( $Z = -6.721, p < .001$ ), and Media Player ( $Z = -6.754, p < .001$ ).

**4.4.5 Intention (Int).** : Friedman ANOVA yielded a statistically significant difference depending on the scenario ( $\chi^2(2) = 98.552, p < .000$ ). Study participants stated to use EMS in Virtual Objects significantly more likely than all other scenarios Motor Learning ( $Z = -4.198, p < .001$ ), Media Player ( $Z = -3.656, p < .001$ ), and Road Safety ( $Z = -7.532, p < .001$ ). While there was no difference between Motor Learning and Media Player, the Road Safety scenario was again rated worst (vs. Motor Learning:  $Z = -5.783, p < .001$ ; vs. Media Player:  $Z = -6.518, p < .001$ ).

**4.4.6 Correlation Analysis.** We also investigated the correlation between the scenarios and users' affinity towards technology. We found a positive co-relation between ATI score and attitude towards Virtual Objects scenario ( $rs(8) = .188, p < .046$ ). Also, there is a negative co-relation between ATI score vs. trust for Road Safety scenario ( $rs(8) = -.280, p < .003$ ) and intent to use for Road Safety scenario ( $rs(8) = -.287, p < .002$ ).

## 5 DISCUSSION

Training process to improve our sensory and motor abilities is not an easy process. To see significant improvement, it is usually necessary to train on a task for an extended period of time with a high number of repetitions in a trial-and-error method. The systems based on EMS have the potential to improve our motor abilities and supplement traditional skill training. It has been demonstrated, for example, that EMS reduces users' response time (RT) when provided prior to their own action (preemptive EMS) [15]. Despite the tendencies that low controllability and safety are associated with less user acceptance, the effect of the other EMS dimensions i.e. intrusiveness and longevity are less clear.

Overall, the following insights can be derived from this experiment with respect to our research question: First, using EMS for VR was rated highest in terms of user acceptance. Participants showed significantly higher attitude towards using VR and also significantly higher intention to use (i.e., the expected behavioral outcome as given by the TAM model) it than all other investigated scenarios. Consequently, EMS is a promising area for VR experiences. The reason could be that haptic feedback is important for immersive VR experiences [19], yet current technology (such as haptic gloves [31]) is limited to certain body parts and can still not provide sensations of heavy resistance. EMS, in contrast, has less limitations, which might allow to provide more fine-grained sensations synchronized over multiple body parts, and has already shown promising results in this context [25]. This was backed by qualitative participant quotes, such as *"This tech could be used in entertainment, mainly for gaming industry [...] in very controlled and restricted fashion."*

In contrast, the Road Safety scenario received the lowest scores – it was rated significantly worse than all other scenarios in all TAM dimensions. Potentially, a reason could be the safety-critical environment, as participants also stated to trust EMS significantly less in this scenario. However, we rather believe this could have emerged from the low controllability, what is supported by participants' statements such as *"I would like to have the*

*complete control all the time"*. Accessing users' body and being able to influence it raises concerns about privacy and security, leading to trust issues amongst potential users. When designing systems incorporating EMS, the users' physical safety should always come first. Further, this view is also backed by the ratings of the other scenario entailing a low "sense of agency" – the Motor Learning scenario. We were surprised that, despite the fact that EMS has the ability to speed up long and intensive learning processes, learning to play the guitar was not universally accepted, and participants expressed strong reservations. Between the EMS technology's usefulness and the discomfort it produces, there is a trade-off. For example, one participant stated *"yes I would like to learn the new instrument faster. At the same time, I may want to do it all by myself. If EMS comes into the picture, I may learn quickly. But may lose on the fun of exploring the learning process."*. Another one reported that *"humans should be able to control machines for better efficiency, and not the other way around except for the fun part"*. Thus, we conclude that scenarios where controllability is in question might be rejected, especially when safety-criticality becomes an additional factor. Even when the technology is exceeding the user's own physical abilities, it shouldn't compromise the sense of agency [40]. In such scenarios, EMS systems must present users with alternatives that allow them keep their sense of agency [41].

Finally, we also cannot confirm that a higher affinity for technology is associated with a higher acceptance of EMS systems. Only 3 of 20 potential correlations (4 scenarios \* 5 TAM dimensions) yielded a statistically significant result. The reason could be that the experience descriptors of EMS are varied and encompass both positive and negative sensations (for example, Hassib et al. [12] have reported descriptions including 'exciting', but also 'hectic', 'unnatural', or 'aggressive'). Future studies will be necessary in this regard. However, we still believe that our findings can be used as a guide to create better user experiences for EMS-based systems.

**Limitations and Future Work.** There are certain limitations to our research. Most importantly, user approval was represented in terms of expected use, as participants were never exposed to the four scenarios due to our research approach (online survey). Second, our EMS taxonomy is simply a first step toward a systematic categorization of prototypes, and its manifestations were expressed using only a limited number of conceivable interaction scenarios. As a result, we must exercise caution when interpreting the effects. Despite the fact that the sample size was large, the majority of the participants were unfamiliar with the technology, and none had ever used it in an interactive context. Future investigations should (1) evaluate more situations and, more crucially, (2) include real-world encounters with EMS prototypes. It would also be interesting to observe whether there's a difference between individuals who have used EMS before and individuals who have never used it before.

## 6 CONCLUSION

Previous studies have mostly focused on the muscle-actuation capability and uses of EMS, with little regard for user acceptance. Instead, we created four scenarios based on a literature study and used the TAM model to measure the technology acceptance of (N=113) participants in an online survey.

Our findings suggest that using EMS to enhance a VR experience is aided by a good attitude and intent to use it. As a result, we recommend that HCI researchers and UX designers leverage EMS in VR to create more engaging and immersive user experiences.

Second, we discovered a rejection of the road safety scenario in both quantitative and qualitative outcomes. As a result, we recommend that HCI researchers and UX designers focus their efforts on keeping users in control of their activities rather than forcing them through EMS. We recommend more research into balancing the benefits of EMS in helping users learn new motor abilities faster while also allowing them to retain control: which

| Md (IQR) |                |                 |              |             |
|----------|----------------|-----------------|--------------|-------------|
| TAM      | Motor Learning | Virtual Objects | Media Player | Road Safety |
| PU       | 5.00 (2.00)    | 5.00 (1.67)     | 5.33 (2.17)  | 3.33 (2.50) |
| PEoU     | 5.20 (1.40)    | 5.00 (1.30)     | 5.00 (1.40)  | 4.40 (1.70) |
| Att      | 5.20 (1.90)    | 5.60 (1.30)     | 5.20 (1.60)  | 3.60 (3.00) |
| trust    | 4.75 (1.25)    | 5.00 (1.50)     | 5.00 (1.50)  | 3.75 (2.50) |
| Int      | 5.00 (3)       | 6.00 (1)        | 5.00 (2)     | 3.00 (4)    |

Table 3: Medians and interquartile range (IQR) of the TAM dimensions in the investigated scenarios

methods, such as "wake words" or interaction modality, are most effective in maintaining control? We believe that trust development and a sense of control are the most crucial elements that systems using this technology should incorporate and explore in order to increase user acceptance and adoption of EMS.

Third, we discovered that using EMS to accelerate learning was not commonly accepted and was outweighed by the desire to maintain control. This emphasizes the necessity of giving people a choice.

In the future, we plan to broaden our study to include people who have experienced EMS in lab trials so that we may compare our scenario-based results to real-life experiences. Another avenue for additional investigation is to include the viewpoints of older participants.

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